

Relax. The semester is nearly over. Choose the *best* answer. Clearly *circle* the letter for your answer. Vague or unclear marks will be counted as incorrect. Each question is worth 2 points. Let's do this thing!

1. Which of the following pairs of structures is *least likely* to represent homology?
 - A. the hemoglobin of a baboon and hemoglobin of a gorilla.
 - B. the brain of a cat and the brain of a dog.
 - C. the wings of a bat and the arms of a human.
 - D. the front flippers of a whale and the front flippers of a seal.
 - E. the wings of a bird and the wings of an insect.
 2. The living great apes consist of the gibbon, the orangutan, and
 - A. baboons.
 - B. hominids
 - C. gorilla and chimpanzee
 - D. chimpanzee and hominids.
 - E. gorilla, chimpanzee, and hominids.
 3. The bones of vertebrate forelimbs are similar because of descent from a common ancestor. These forelimbs can be called _____ structures.
 - A. similar
 - B. intelligently designed
 - C. analogous
 - D. radius and ulna
 - E. homologous
 4. The process of evolution occurs by
 - A. a combination of mutation and random natural selection.
 - B. by removing the weak individuals and leaving the strong individuals.
 - C. entirely random processes.
 - D. a combination of random mutation and natural selection.
 - E. the environment filtering out less fit individuals.
 5. Of the species listed below, the oldest (farthest back in time) ancestor of humans is commonly thought to be
 - A. *Australopithecus africanus*.
 - B. *Australopithecus boisei*.
 - C. *Australopithecus afarensis*.
 - D. *Homo habilis*.
 - E. *Homo erectus*.
-

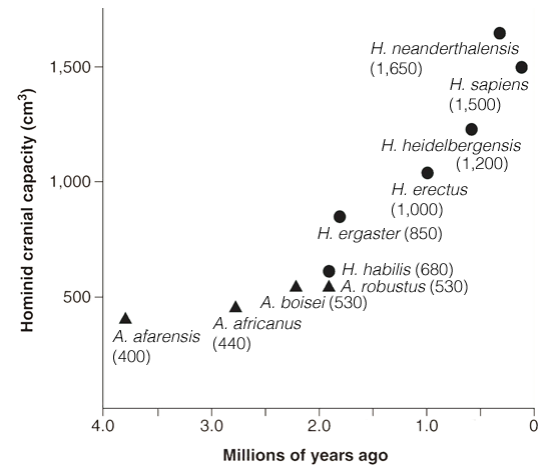
6. Genetic changes that occur in organisms through time can help us
 - A. determine relatedness of organisms.
 - B. understand how groups of organisms evolved over time.
 - C. understand how individual traits evolved, such as wings from arms.
 - D. construct molecular family trees.
 - E. all of these answers are possible.
 7. Choose the phrase that best describes evolutionary change.
 - A. One species evolves into another species by genetic changes.
 - B. Genetic changes in a population over time.
 - C. Genetic changes in a population.
 - D. New kinds of animals replace older kinds of animals over time.
 - E. Genetic changes in a species over time.
 8. Which of the following is not an observation or inference on which natural selection is based?
 - A. Individuals whose traits are best suited to the environment generally leave more offspring than those whose traits are less suited.
 - B. Poorly adapted individuals never produce offspring.
 - C. Species produce more offspring than the environment can support.
 - D. Only a fraction of the offspring produced by an individual may survive.
 - E. There is genetic variation among individuals that can be inherited.
 9. Dolphins and fishes have streamlined bodies for swimming through water, which is more dense than air. Similar structures, such as streamlined bodies, that are necessary to serve the same type of functions are called
 - A. evolutionary innovations.
 - B. adaptations.
 - C. analogies.
 - D. evolutionary traits.
 - E. homologies.
 10. Fitness is measured as
 - A. the relative strength or weakness of an individual.
 - B. the number of offspring produced over the life time of an individual.
 - C. the number of offspring produced in a single reproductive event.
 - D. the relative number of offspring produced in a community.
 - E. the total number of offspring produced in a population.
-

11. Which fish has the greatest total fitness? (A clutch is the result of one breeding event.)

- A. One that lives for 3 years, has 4 clutches per year, and 50 offspring per clutch.
- B. One that lives for 4 years, has 3 clutches per year, and 45 offspring per clutch.
- C. One that lives for 3 years, has 2 clutches per year, and 72 offspring per clutch.
- D. One that lives for 3 years, has 3 clutches per year, and 60 offspring per clutch.
- E. One that lives for 2 years, has 3 clutches per year, and 50 offspring per clutch.

12. The figure at right shows that

- A. Brain capacity had to be at least 1500 cubic centimeters (cm^3) before language could evolve.
- B. bipedalism evolved before large brain size.
- C. *Homo habilis* was probably smarter than *Australopithecus robustus*.
- D. Over two million years, brain size in *Homo* increased by about 1000 cubic centimeters (cm^3).
- E. *Homo neanderthalensis* was probably smarter than *Homo sapiens*.



13. Traits in an organism that increase fitness are called

- A. adaptations.
- B. analogies.
- C. homologies.
- D. phenotypes.
- E. genes.

14. DNA, chromosomes and fossil evidence support the hypothesis that *Homo sapiens* originated in

- A. western Europe
- B. many different regions that were first occupied by *Homo erectus*.
- C. eastern Africa
- D. western Africa
- E. eastern Asia

15. Complex language probably evolved in Cro-magnons. Which species of *Homo* evolved just before Cro-magnons?

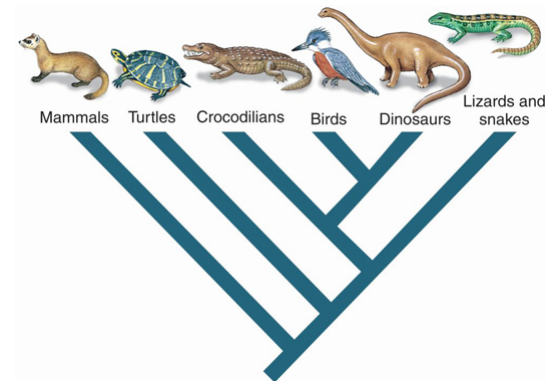
- A. *Homo sapiens*
- B. *Australopithecus afarensis*
- C. *Homo habilis*
- D. *Homo rudolfensis*
- E. *Homo erectus*

16. The first species to use symbolic language is

- A. *Homo erectus*.
- B. *Homo neanderthalensis*.
- C. *Homo ergaster*.
- D. *Homo habilis*.
- E. *Homo sapiens*.

17. According to this phylogenetic tree,

- A. birds are equally related to dinosaurs and crocodilians.
- B. birds are more closely related to dinosaurs than to mammals.
- C. turtles are equally related to lizards and mammals.
- D. mammals are more closely related to turtles than to crocodilians.
- E. dinosaurs are extinct.



18. The first hominoid species to use stone tools was probably

- A. Java man.
- B. Peking man.
- C. *Australopithecus afarensis*.
- D. *Homo habilis*.
- E. *Homo erectus*.

19. All of the following anatomical characteristics of *Homo* contributed to bipedalism except

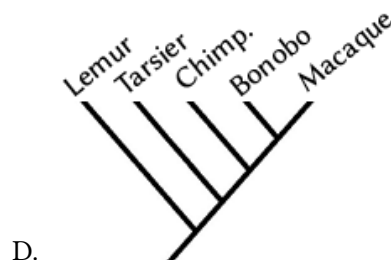
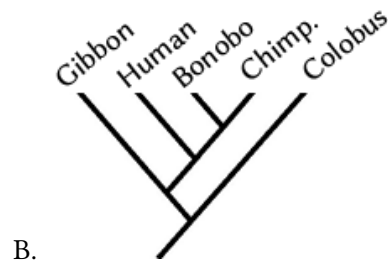
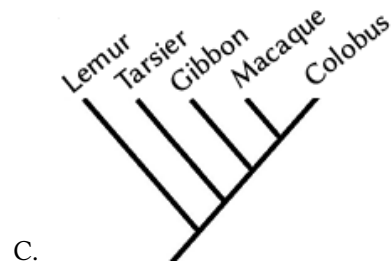
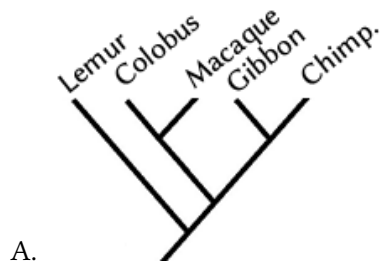
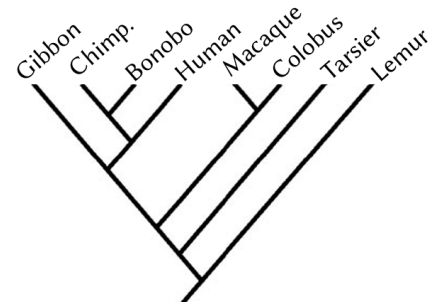
- A. longer lower limbs.
- B. increased brain capacity.
- C. hole at the base of the skull for the spinal cord.
- D. bowl-shaped pelvis.
- E. curved vertebral column.

20. All of the following are part of the process of natural selection except

- A. All populations overproduce offspring.
- B. Individuals compete for limited resources.
- C. Traits that increase fitness do not change in prevalence over time.
- D. Genetic variation is present in a population.
- E. Not all offspring produced will survive.

21. Christina has many food allergies, so she has to be careful about unknown foods. Christina stayed with her friend Sarah during the winter break. For dinner one evening, Sarah made a salad that included raw starfruit. Christina had never had starfruit before. On the internet, Christina learned that starfruit belongs to the family Oxalidales. Based on the phylogeny presented in the figure *on screen*, should Christina try starfruit? Choose the best answer.
- A. No. Oxalidales shares a common ancestor with other vegetables that are not safe to eat.
 - B. No. Oxalidales is a rosid. All rosids cause allergic reactions for Christina.
 - C. Yes. Some rosids are safe to eat. Starfruit should also be safe to eat.
 - D. Yes. Oxalidales is most closely related to other rosids that are safe to eat.
 - E. Yes. Oxalidales is in the eudicots group, most of which are safe to eat.

22. Small phylogenetic trees are representatives of larger phylogenetic trees. The small trees must convey the same relationships (same ancestors) given by the larger tree. The tree at right shows the relationships among various primates. Which of the following small trees shows a *different* relationship from the primate tree?



23. Female guppies prefer to mate with brightly colored male guppies, and are common in pools without predators. However, in pools with predators, males have drab colors. Which of the following is the best evolutionary explanation for this observation?
- A. Drab males are usually faster than brightly colored males and so are harder for predators to catch.
 - B. Drab males have higher fitness than brightly colored males in pools without predators.
 - C. Brightly colored males are always slower than drab males so are more easily captured by predators.
 - D. Predators prefer the taste of brightly colored males.
 - E. Drab males have higher fitness than brightly colored males in pools with predators.

24. One characteristic that scientists use to tell if a fossil is *Homo habilis* or *Homo erectus* is
- A. hole at the base of the skull for the spinal cord.
 - B. brain size.
 - C. presence in Africa.
 - D. walking upright.
 - E. opposable thumbs.
25. Which hypothesis argues that *Homo sapiens* evolved in Africa and colonized Europe and Asia in at least two different migratory events, separated by about 100,000 years?
- A. Recently Out of Africa model.
 - B. Asian-African hypothesis.
 - C. Local hypothesis.
 - D. Multiregional hypothesis.
 - E. Multimigratory Hypothesis
26. Flounders are unique fishes that lay on one side instead of swimming upright. Also unlike all other fishes, the flounder's eyes are on the same side of their head. If flounders share a common ancestor with other fishes, then which of the following is the best prediction about the early development of flounders?
- A. After they hatch, flounders will lay on their side and have both eyes on the same side of their head.
 - B. After they hatch, flounders will have one eye on each side of its head, and swim upright like other fishes.
 - C. After they hatch, flounders will lay on their side but have one eye on each side of its head.
 - D. Flounders probably didn't share a common ancestor with other fishes.
 - E. After they hatch, flounders will swim upright but have both eyes on the same side of its head.
- 
27. Several species of warblers can live in the same spruce tree. They can co-exist in the same tree because
- A. they evolved adaptations that other species of birds, like sparrows, did not.
 - B. some species of warblers evolved to fly higher in the tree than others species of warblers.
 - C. some species of warblers have greater fitness than other species of warblers.
 - D. the warblers evolved adaptations that allow them to occupy separate niches, which reduced competition for limited resources.
 - E. they all evolved better fitness.

28. A group of organisms related by descent and common ancestry is
- A. homologous.
 - B. a clade.
 - C. an evolutionary tree.
 - D. a phylogenetic tree.
 - E. a family.
29. All of the following can result from or are part of the process of natural selection except
- A. populations can evolve adaptive traits over time.
 - B. traits that increase fitness become more common in a population over many generations.
 - C. beneficial traits become adaptations.
 - D. some individuals will survive longer than other individuals in a population.
 - E. an individual can adapt to become a better competitor.
30. Which feature appeared first in early human ancestors?
- A. language.
 - B. tool use.
 - C. cave drawings.
 - D. bipedalism.
 - E. increased brain size.
-